

Machine Dreaming

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Figure 1: Interacting with *Machine Dreaming*

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → *Interaction design process and methods*; • **Computing methodologies** → *Artificial intelligence*.

ACM Reference Format:

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1 Abstract

Machine Dreaming is an interactive installation that explores the perceptual and social dynamic between humans and AI, highlighting the tensions and intimacies that surface in the experience of being ‘seen’ and perceived by these systems. The work offers a space where people can explore their relationship to AI through visual, gestural, and movement-based interaction. The work reveals how sensory ambiguity and aesthetic transformation can surface deeper emotional responses in relation with AI.

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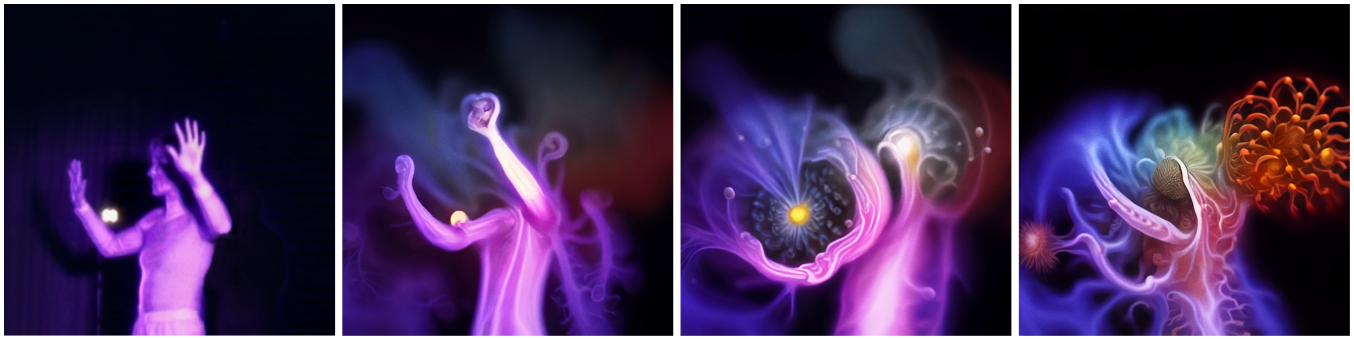


Figure 2: A transformation sequence in *Machine Dreaming*

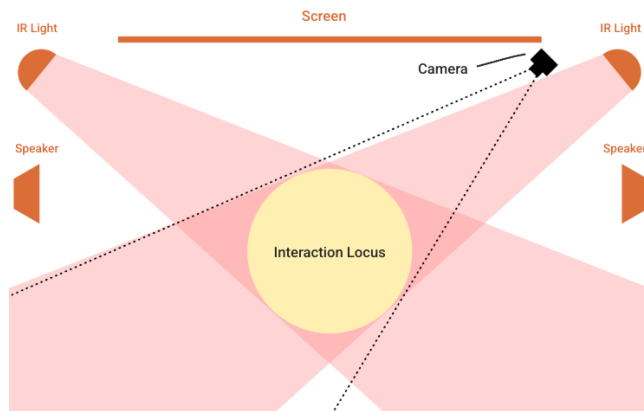


Figure 3: Top down view of installation space

Machine Dreaming is set in a darkened room, with a dim central spotlight used to indicate the interaction locus (Figure 3). The interactive installation uses a real-time video feed that captures viewers' presence in the space. The feed is processed through a custom-built system in TouchDesigner¹, incorporating MediaPipe² for gesture recognition and Stable Diffusion³ for image generation. The projected image of the viewer is gradually transformed through generative AI, producing a shifting representation of the viewer from human to alien-like plant forms, responding to their movements in real time. The work operates as a continuous feedback loop: the viewer's movements shape the generative output, which in turn subtly guides their behaviour. This dynamic facilitates a sense of embodiment and presence, helping viewers to perceive their interaction with the system as relational rather than passive.

Slowness is intentionally built into the interaction. Subtle movements cause the work to produce more vivid transformations, deeper sound, and increasingly intricate generative visuals. This creates a responsive, embodied feedback loop where presence, intention and generative AI co-determine the overall aesthetic. As viewers slow their movements, the work unfolds fully, the live video image progressively transforms into the AI's interpretation constrained by prompts (i.e. botanical forms, spatial depth, and aesthetic detail) and training data. In the final stage of the work, botanical forms appear to grow from the viewer's form, beginning to animate and extend further into the surrounding space. In this process, participants encounter themselves as fluid, data-driven forms, refracted through the training sets of the model. The work invites viewers to experience being both subject and co-creator within a relational perceptual system, prioritising embodied meaning-making through gesture and interaction over cognitive explanation.

By foregrounding this interplay between human presence and algorithmic interpretation, *Machine Dreaming* highlights the conditions of being seen and perceived by non-human intelligence. The work prompts viewers to reflect on their relationship to AI and how perception, agency, and representation are negotiated with these systems.

¹<https://derivative.ca/>

²<https://github.com/torinmb/mediapipe-touchdesigner>

³Via the StreamDiffusion API: <https://github.com/cumulo-autumn/StreamDiffusion?tab=readme-ov-file>

Acknowledgments

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